



THESIS ASSIGNMENT

Name and Surname: Jakub Novotný
Study programme: Data Science (Joint degree study, bachelor I. deg., full time form)
Field of Study: Computer Science
Mathematics
Type of Thesis: Bachelor's thesis
Language of Thesis: English
Secondary language: Slovak

Title: Machine Learning Assessment of the Trading Value of Central Bank Information

Annotation: The thesis will examine how financial markets absorb and price information from official central bank communications. It will apply Natural Language Processing techniques and machine learning models to textual datasets, focusing on ECB press conferences. The empirical work will extract monetary policy signals and sentiment from these texts. The goal is to test different approaches to data preprocessing, topic classification, and contextual analysis. The thesis will trace how extracted signals evolve over time and interact with key macroeconomic and market variables and measures of market uncertainty. It will evaluate how well different communication formats and phases predict market events and responses. It will also assess the strengths and limitations of various machine learning architectures for financial and economic text analysis.

Beyond its academic contribution, the thesis will lay the groundwork for an NBS research project on the informational value of central bank communications. Successful results may feed directly into policy-relevant research and potentially lead to a co-authored publication with NBS.

Supervisor: Ing. Pavel Gertler, PhD.
Department: FMFI.KI - Department of Computer Science
Head of department: prof. RNDr. Martin Škoviera, PhD.

Assigned: 30.10.2025

Approved: 04.11.2025 doc. Mgr. Tomáš Vinař, PhD.
Guarantor of Study Programme

Student

Supervisor